

Family 7.1: ocean colour joins the global tensor

a fourth channel group, inherited rather than rebuilt

blauewelt/earth — design note

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Abstract

Family 7.1 extends the global input tensor of family 7 — the Earth’s surface state on a 0.25° grid from pole to pole in five-day bins, 1982–2024, in 54 channels — with a fourth channel group carrying the observed surface chlorophyll-a concentration from the ESA Ocean Colour Climate Change Initiative (4km, September 1997 to December 2024). The group holds two channels: the block mean of $\log_{10}$ chlorophyll over the 36 source cells of each 0.25° point, and the fraction of cell-days in the pentad that carried a retrieval. Its time axis starts where the record starts, so the group is 1,997 bins rather than 3,142. The three inherited groups are not recomputed: the build hard-links the published family-7 files, refuses any write through the new names, and its manifest states that the inherited files hash identically to family 7’s. We give the source, the averaging rule, the missing-value semantics, the layout, the assertions, and the reason the channel is a target before it is an input.

1 Purpose

The programme’s forward model reads a per-cell state, predicts it forward, and reads out quantities of interest from the predicted state. Surface colour — the chlorophyll-a concentration retrieved from water-leaving radiance — is the only global, routine observation of the ocean biosphere, and it is one of the read-outs the programme has committed to (E-072). A read-out needs a target on the tensor’s own grid and time axis; family 7 had none. Family 7.1 adds one.

Two decisions frame the addition. First, colour enters as a *target* and probe before it enters as an input: the question it is built to answer is whether the physical state the model already carries — temperature, mixed layer, currents, wind, light through the calendar — predicts the biosphere a season ahead. Second, the source is averaged onto the 0.25° grid rather than kept at its native 4km. That is a deliberate and temporary simplification: it lets the channel drop into the three-group layout with no new machinery, at the price of discarding a factor of 36 in pixels. The companion note on family 10 describes the layout in which fine sources are kept at their own granularity; family 7.1 is a tier-G group of that design, unchanged.

2 Source

ESA Ocean Colour CCI version 6.0, geographic projection, daily, chlorophyll-a only [1]: the merged SeaWiFS, MERIS, MODIS-Aqua, VIIRS and OLCI record, bias-corrected across sensors, on a $1/24^\circ$ grid of 4320×8640 cells with centres at odd multiples of $1/48^\circ$, 4 September 1997 to 31 December 2024. The variable is `chlora` in mg m^{-3} , float32 with a fill value. The retrieval exists only where the sensor saw the water: not under cloud, sun glint or sea ice, not in polar night, not over land. The builder does not assume file names; it lists each year’s directory on the archive’s own catalogue and preflights one real file before spending anything. The daily uncertainty fields shipped with the product are not stored — an uncertainty of a merged algorithm is not an observation — and the gap-filled level-4 products are not used, because the target should be what was seen.

#	channel	unit	rule
0	log_chl	$\log_{10} \text{ mg m}^{-3}$	per day, the mean of $\log_{10}$ chlorophyll over the finite cells of the 6×6 block centred on the point; per pentad, the mean of those daily means over the days with at least one finite cell; absent if no day had one
1	chl_cov	fraction in (0, 1]	finite cell-days in the block during the pentad, divided by (cells in the block $\times$ days in the bin); absent exactly where log_chl is

Table 1: Group `oc025`: two ocean-colour channels at 0.25° , shape `[1997, 721, 1440, 2]`, z-scored per channel like every other group, the standardisation constants stored as `norm_oc025`.

3 Channels

The block. A point of the 0.25° grid sits at a multiple of $0.25^\circ = 6/24^\circ$, which is a cell boundary of the $1/24^\circ$ grid; the six cells whose centres lie within $\pm 0.125^\circ$ of the point on each axis are therefore exactly three on either side. The block wraps in longitude; at the two pole rows it is truncated to the cells that exist. The implementation is a roll by three cells and a reshape, so the pole and the dateline are the only special cases and both are exercised by tests.

Log space. Chlorophyll is log-normal over four orders of magnitude. The arithmetic mean of a block containing one bloom cell at 30 mg m^{-3} and thirty-five at 0.05 describes neither; the mean of $\log_{10}$ does. The programme’s read-out for colour is scored in log space (E-072), so the stored quantity and the scored quantity agree.

Coverage. A pentad mean resting on one clear pixel on one day and one resting on thirty-six pixels on five days are different amounts of evidence, and the model should be told which it is looking at. `chl_cov` carries that. It is stored un-logged and z-scored like every other channel; its distribution is a property of the observing system (clear skies at low latitude, cloud and polar night at high) and is reported from the built file rather than assumed here.

4 Missing values

`NaN` keeps its family-7 meaning — not observed here at this time — and for colour it is the rule rather than the exception: at high latitudes in winter no day is observed for months. Before bin 1145, the bin that contains 4 September 1997, there is no observation at all, and the group’s time axis starts there. `chl_cov` is absent exactly where `log_chl` is; a block that saw nothing has no coverage to report, and storing zero there would collapse the distinction between “unobserved” and “observed to be empty” that the missing-token design relies on. Nothing is filled: no climatological fill, no interpolation across days, no level-4 product.

5 Layout and inheritance

The group is one file, `family7_global025_pentad_l1.X_oc025.npy`: float16, C order, a 128-byte NumPy header, bin-major so that one pentad is a contiguous 4.15 MB range read. Its first axis is offset — row r holds bin $1145 + r$ — and the offset is stored in the metadata file as `oc_bin_first` beside an explicit bin index. The offset saves 4.7 GB of empty slabs (8.3 GB instead of 12.9); it costs a per-group first bin in every consumer, which the samplers and the front end already supported for the monthly Argo group.

Family 7.1 is a new recipe, `f711`, with its own stem and its own folder on the public dataset repository, because the family-7 files, their hashes and their documentation are cited elsewhere and a published dataset should not change under its own name. The three inherited groups are *hard-linked* from the finished family-7 build directory — one inode, no extra bytes — and never recomputed. Since a hard link shares its inode, a write through the new name would corrupt the published bytes, and the process runs as root, which ignores file modes; so the refusal is structural rather than a permission bit: every writable open of a group file and the stage machinery’s discard path check whether the group was seeded and exit if so. The publish step fetches family 7’s manifest and asserts that each inherited file

hashes identically before it writes `same_as_f7l0: true` beside that file’s entry. The metadata file is regenerated and every key it shares with family 7’s is required to be equal.

6 Assertions

Before the tensor is trusted: the inherited files are byte-identical to family 7’s manifest; the group’s shape, dtype and header length are as stated; the first bin is computed from the epoch, not typed; every finite coverage lies in $(0, 1]$ and `log_chl` is finite exactly where coverage is; the block mean reproduces exact values on synthetic input (a block of $10^{0.5}$ on all five days reads 0.5 with coverage 1; one finite cell on one day reads its own $\log_{10}$ with coverage $1/180$); the pole rows and the dateline are exercised; the stage resumes across a simulated interruption; and on the bin opening 3 January 2015 — the comparison bin of the family-7 validation — the global mean of `log_chl` and the fraction of observed ocean cells are written to the build record for comparison, by eye, against the chlorophyll layers already on the globe.

7 Holdout

Unchanged: 2009, 2017 and 2023 are development holdout years; the terminal split trains through 2020 and tests on 2021–2024. Colour’s record begins in September 1997, so a training-years-only seasonal climatology for the channel has twenty-one years to fit from.

8 Limitations

The averaging discards 35 of every 36 source pixels; the coverage channel records the loss but does not undo it. The channel is clear-sky and daylight only, which halves its live bins at high latitude and removes the polar winter entirely. Reflectance bands, which the programme’s rule names as the input from which colour is the target, are not carried; they belong to the first design that keeps fine sources at their own granularity. And the record is 27 years against the tensor’s 43.

References

- [1] Sathyendranath, S. et al. (2019). An Ocean-Colour Time Series for Use in Climate Studies: The Experience of the Ocean-Colour Climate Change Initiative (OC-CCI). *Sensors* 19(19), 4285. Version 6.0 data: ESA Climate Office / Plymouth Marine Laboratory.
- [2] blauewelt/earth (2026). Family 7: a global, three-resolution input tensor for a forward model of the Earth’s surface state. Design note, 7 September 2026.